

Solar Si appendix

Sun · Si · downloadable appendix

Solar silicon: 7.489 ± 0.074

Asplund, Amarsi & Grevesse 2021: 7.51 ± 0.03 · this measurement -0.021 dex

Kitt Peak · VIS · Reference Grade · SET-SI_AGSS21 · ENGINE-A · n = 7. The top-ranked admitted product (Reference Grade, then the most complete treatment, then line count, then total uncertainty); independent products and engines are not averaged together.

Highlighted Products

VIS

7.489 ± 0.074

± 0.020 stat · ± 0.071 syst

SET-SI_AGSS21 · Si I · Synth · 1D-NLTE · Amarsi · SYNTH · ENGINE-A · line set not supplied · solar_kpno_molecfits_corrected · Reference Grade

red-optical

7.578 ± 0.127

± 0.035 stat · ± 0.122 syst

SET-SI_AGSS21 · Si I · Synth · 1D-NLTE · Amarsi · SYNTH · ENGINE-A · line set not supplied · solar_kpno_molecfits_corrected · Reference Grade

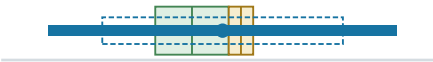
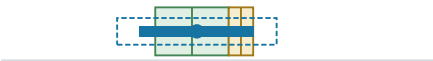
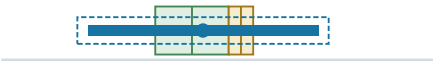
NIR

7.549 ± 0.086

± 0.030 stat · ± 0.081 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-NLTE · Amarsi · SYNTH · ENGINE-A · line set not supplied · solar_kpno_molecfits_corrected · Reference Grade

Error-bar Forest Plot

VIS — harps solar_harps_molecfits_corrected A(Si) dex; axis 7.35–7.71		
SET-SI_AGSS21 · Si II · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=1		7.535 ± 0.142 stat ± 0.098 syst
SET-SI_AGSS21 · Si I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=7		7.514 ± 0.047 stat ± 0.065 syst
SET-SI_AGSS21 · Si I · Synth · 1D-NLTE · Amarsi · SYNTH · ENGINE-A · line set not supplied Reference Grade · ALL · n=7		7.487 ± 0.048 stat ± 0.072 syst
VIS — iag_fts_solar_atlas solar_iag A(Si) dex; axis 7.35–7.71		
SET-SI_AGSS21 · Si II · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied Reference Grade · ALL · n=1		7.519 ± 0.094 stat ± 0.103 syst

VIS — iag_fts_solar_atlas solar_iag A(Si) dex; axis 7.35-7.71

SET-SI_AGSS21 · Si I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Reference Grade · ALL · n=7



7.511
±0.015 stat ±0.070 syst

SET-SI_AGSS21 · Si I · Synth · 1D-NLTE · Amarsi ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=7



7.484
±0.016 stat ±0.076 syst

VIS — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(Si) dex; axis 7.35-7.71

SET-SI_AGSS21 · Si II · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Reference Grade · ALL · n=1



7.528
±0.144 stat ±0.102 syst

SET-SI_AGSS21 · Si I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Reference Grade · ALL · n=7



7.513
±0.020 stat ±0.065 syst

SET-SI_AGSS21 · Si I · Synth · 1D-NLTE · Amarsi ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=7



7.486
±0.021 stat ±0.071 syst

VIS — kpno_solar_atlas solar_kpno_molecfit_corrected A(Si) dex; axis 7.35-7.71

SET-SI_AGSS21 · Si II · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Reference Grade · ALL · n=1



7.518
±0.110 stat ±0.110 syst

SET-SI_AGSS21 · Si I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Reference Grade · ALL · n=7



7.516
±0.019 stat ±0.064 syst

SET-SI_AGSS21 · Si I · Synth · 1D-NLTE · Amarsi ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=7



7.489
±0.020 stat ±0.071 syst

VIS axis: 7.35 · 7.44 · 7.53 · 7.62 · 7.71 A(Si) dex

red-optical — iag_fts_solar_atlas solar_iag A(Si) dex; axis 7.41-7.74

SET-SI_AGSS21 · Si I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Reference Grade · ALL · n=2



7.591
±0.030 stat ±0.121 syst

SET-SI_AGSS21 · Si I · Synth · 1D-NLTE · Amarsi ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=2



7.573
±0.028 stat ±0.125 syst

red-optical — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(Si) dex; axis 7.41-7.74

SET-SI_AGSS21 · Si I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Reference Grade · ALL · n=2



7.581
±0.035 stat ±0.120 syst

SET-SI_AGSS21 · Si I · Synth · 1D-NLTE · Amarsi ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=2



7.563
±0.033 stat ±0.123 syst

red-optical — kpno_solar_atlas solar_kpno_molecfits_corrected A(Si) dex; axis 7.41-7.74

SET-SI_AGSS21 · Si I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Reference Grade · ALL · n=2



7.596
±0.037 stat ±0.118 syst

SET-SI_AGSS21 · Si I · Synth · 1D-NLTE · Amarsi ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=2



7.578
±0.035 stat ±0.122 syst

red-optical axis: 7.41 · 7.49 · 7.58 · 7.66 · 7.74 A(Si) dex

NIR — crires_plus solar_crires_plus_j_rya1219 A(Si) dex; axis 7.26-7.73

SET-SI_DESHMUKH2022 · Si I · Synth · 1D-LTE ·
SYNTH · 1D-LTE · line set not supplied
Reference Grade · ALL · n=2



7.513
±0.005 stat ±0.086 syst

SET-SI_DESHMUKH2022 · Si I · Synth · 1D-NLTE ·
Amarsi · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=2



7.487
±0.009 stat ±0.091 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-LTE · SYNTH
· 1D-LTE · line set not supplied
Reference Grade · ALL · n=9



7.518
±0.031 stat ±0.084 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-NLTE ·
Amarsi · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=9



7.496
±0.030 stat ±0.089 syst

NIR — crires_plus solar_crires_plus_y_rya794 A(Si) dex; axis 7.26-7.73

SET-SI_DESHMUKH2022 · Si I · Synth · 1D-LTE ·
SYNTH · 1D-LTE · line set not supplied
Reference Grade · ALL · n=1



7.468
±0.018 stat ±0.135 syst

SET-SI_DESHMUKH2022 · Si I · Synth · 1D-NLTE ·
Amarsi · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=1



7.428
±0.018 stat ±0.139 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-LTE · SYNTH
· 1D-LTE · line set not supplied
Reference Grade · ALL · n=2



7.470
±0.002 stat ±0.096 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-NLTE ·
Amarsi · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=2



7.442
±0.015 stat ±0.101 syst

NIR — crires_plus solar_crires_plus_y_wide_rya1054 A(Si) dex; axis 7.26-7.73

SET-SI_DESHMUKH2022 · Si I · Synth · 1D-LTE ·
SYNTH · 1D-LTE · line set not supplied
Reference Grade · ALL · n=1



7.468
±0.018 stat ±0.135 syst

SET-SI_DESHMUKH2022 · Si I · Synth · 1D-NLTE ·
Amarsi · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=1



7.428
±0.018 stat ±0.139 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-LTE · SYNTH
· 1D-LTE · line set not supplied
Reference Grade · ALL · n=5



7.471
±0.020 stat ±0.065 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-NLTE ·
Amarsi · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=5



7.458
±0.021 stat ±0.072 syst

NIR — iag_fts_solar_atlas solar_iag A(Si) dex; axis 7.26-7.73

SET-SI_DESHMUKH2022 · Si I · Synth · 1D-LTE ·
SYNTH · 1D-LTE · line set not supplied
Reference Grade · ALL · n=1



7.491
±0.092 stat ±0.136 syst

SET-SI_DESHMUKH2022 · Si I · Synth · 1D-NLTE ·
Amarsi · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=1



7.451
±0.092 stat ±0.139 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-LTE · SYNTH
· 1D-LTE · line set not supplied
Reference Grade · ALL · n=4



7.527
±0.018 stat ±0.140 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-NLTE ·
Amarsi · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=4



7.515
±0.023 stat ±0.143 syst

NIR — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(Si) dex; axis 7.26-7.73

SET-SI_ELGUETA2026 · Si I · Synth · 1D-LTE · SYNTH
· 1D-LTE · line set not supplied
Reference Grade · ALL · n=1



7.558
±0.098 stat ±0.145 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-NLTE ·
Amarsi · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=1



7.546
±0.098 stat ±0.148 syst

NIR — kpno_solar_atlas solar_kpno_molecfits_corrected A(Si) dex; axis 7.26-7.73

SET-SI_DESHMUKH2022 · Si I · Synth · 1D-LTE ·
SYNTH · 1D-LTE · line set not supplied
Reference Grade · ALL · n=3



7.581
±0.030 stat ±0.069 syst

SET-SI_DESHMUKH2022 · Si I · Synth · 1D-NLTE ·
Amarsi · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=3



7.552
±0.035 stat ±0.075 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-LTE · SYNTH
· 1D-LTE · line set not supplied
Reference Grade · ALL · n=15



7.556
±0.030 stat ±0.075 syst

SET-SI_ELGUETA2026 · Si I · Synth · 1D-NLTE ·
Amarsi · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=15



7.549
±0.030 stat ±0.081 syst

NIR axis: 7.26 · 7.38 · 7.50 · 7.61 · 7.73 A(Si) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every silicon value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the silicon abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Solar silicon context

Silicon is traced by Si I lines from the blue to the H band and by the single Si II 6371 Å line. The optical lines are Asplund et al. 2021's set (Amarsi & Asplund 2017) with the gf the authors used; the infrared lines are the published sets of Deshmukh et al. 2022 and Elgueta et al. 2026 on Pehlivan Rhodin et al. 2024 oscillator strengths. Each set, spectrum and treatment remains a separate product.

[Published line sets](#) · [gf uncertainty model](#) · [Methodology](#)

Download the evidence

[Download PDF — Solar Si appendix](#)

[Source feed — includes quarantined fit outputs, not adopted abundances](#)

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

Reproducibility

Generated from the pinned science snapshot. Source commit c971602d5bc290c1a5567b4c8d8b809cded46ced
Feed Si.json v1.136 · updated 2026-10-09T00:18:06Z

Generated 2026-10-08T18:18:29-06:00 (source commit timestamp)

Feed SHA-256 09a86f12203c14c1099fa2714e8bbc18d92e739a7b1440a0d78fff52fb25b074

Generator scripts/generate_cno_publication.py v1.0.0

[Pinned scientific source](#)

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Si comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Solar abundance and indicator analyses. Published line set SI_AGSS21 (the line selection). Sources: [3].

Atomic data and gf authorities. Si II oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3], [4], [5].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [6].

Solar abundance and indicator analyses. Published line set SI_DESHMUKH2022 (the line selection). Sources: [7].

Atomic data and gf authorities. Si I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [8].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Si_Amarsi2020_PySME.csv). Sources: [9].

Solar abundance and indicator analyses. Published line set SI_ELGUETA2026 (the line selection). Sources: [10].

Product-to-source index

Si-products-18313ebf6e74 · products · VIS · solar_harps_molecfits_corrected · SET-SI_AGSS21 · 1D-LTE · not supplied · [1], [2], [3], [4], [5], [6]

Si-products-7155cb74f406 · products · VIS · solar_iag · SET-SI_AGSS21 · 1D-LTE · not supplied · [1], [2], [3], [4], [5], [6]

Si-products-7b250bcd5f9f · products · VIS · solar_kpno_kurucz2005_corrected · SET-SI_AGSS21 · 1D-LTE · not supplied · [1], [2], [3], [4], [5], [6]

Si-products-f0069c9e7ce2 · products · VIS · solar_kpno_molecfits_corrected · SET-SI_AGSS21 · 1D-LTE · not supplied · [1], [2], [3], [4], [5], [6]

Si-products-01c39c8b42fa · products · NIR · solar_crires_plus_j_rya1219 · SET-SI_DESHMUKH2022 · 1D-LTE · not supplied · [1], [2], [7], [8], [6]

Si-products-72be3818e99d · products · NIR · solar_crires_plus_j_rya1219 · SET-SI_DESHMUKH2022 · ENGINE-A · not supplied · [1], [2], [7], [8], [9]

Si-products-c0bc9e0f8915 · products · NIR · solar_crires_plus_j_rya1219 · SET-SI_ELGUETA2026 · 1D-LTE · not supplied · [1], [2], [10], [8], [6]

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Si-products-96afd80cbd07 · products · NIR · solar_crires_plus_y_rya794 · SET-SI_DESHMUKH2022 · 1D-LTE · not supplied · [1], [2], [7], [8], [6]

Si-products-50abd0490f95 · products · NIR · solar_crires_plus_y_rya794 · SET-SI_DESHMUKH2022 · ENGINE-A · not supplied · [1], [2], [7], [8], [9]

Si-products-1af1f5a06e40 · products · NIR · solar_crires_plus_y_rya794 · SET-SI_ELGUETA2026 · 1D-LTE · not supplied · [1], [2], [10], [8], [6]

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Si-products-89a2efb973b7 · products · NIR · solar_crires_plus_y_wide_rya1054 · SET-SI_ELGUETA2026 · ENGINE-A · not supplied · [1], [2], [10], [8], [9]

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Si-products-a20959b10970 · products · NIR · solar_iag · SET-SI_ELGUETA2026 · 1D-LTE · not supplied · [1], [2], [10], [8], [6]

Si-products-5017fab6a292 · products · NIR · solar_iag · SET-SI_ELGUETA2026 · ENGINE-A · not supplied · [\[1\]](#), [\[2\]](#), [\[10\]](#), [\[8\]](#), [\[9\]](#)

Si-products-403380bb313c · products · NIR · solar_kpno_kurucz2005_corrected · SET-SI_ELGUETA2026 · 1D-LTE · not supplied · [\[1\]](#), [\[2\]](#), [\[10\]](#), [\[8\]](#), [\[6\]](#)

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Si-products-46cc8e0649e9 · products · VIS · solar_harps_molecfite_corrected · SET-SI_AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

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Si-products-389584e88ad2 · products · VIS · solar_kpno_kurucz2005_corrected · SET-SI_AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#)

Si-products-c8d81b5515f4 · products · VIS · solar_kpno_kurucz2005_corrected · SET-SI_AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

Si-products-372f6baf6f78 · products · VIS · solar_kpno_molecfite_corrected · SET-SI_AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#)

Si-products-a3036e85f010 · products · VIS · solar_kpno_molecfite_corrected · SET-SI_AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

Si-products-743145529f54 · products · red-optical · solar_iag · SET-SI_AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[6\]](#)

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Complete applicable bibliography

Solar abundance and indicator analyses

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